When Verbs Forget

Evidence for Negatively Accelerated Decay in Sentence Processing

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INTRODUCTION

Background

- **Assumption: Linguistic representations decay non-linearly** (steep initial drop, then leveling off):
  - Sigmoidal/logarithmic (Gibson 1998; Grodner & Gibson 2005)
  - Power law (Lewis & Vasishth 2005; ACT-R)
- **Issue: Evidence comes from general memory research**, which may not reflect sentence processing.
- **Open question: Do linguistic representations indeed decay in negatively accelerated fashion?**

Design logic

- **Verbs impose selectional restrictions on their complements.** DEPEND is acceptable with a PP[*on*], but less so with a CP:
  - *The project depends* [*on timely decisions*].
  - ? *The project depends* [*that the morale stays high*].
- **As these restrictions decay,** increasing the distance between the verb and the problematic complement may improve acceptability (Kim & Lu 2024).
  - ? *The project depends on stakeholder input, and* [*that the morale stays high*].
- **If decay is negatively accelerated,** improvement should eventually plateau:
  - ? *The project depends* [*that the morale stays high*].
  - ? *The project depends on stakeholder input, and* [*that the morale stays high*].
  - ? *The project depends on stakeholder input, on timely decisions, and* [*that the morale stays high*].

METHODOLOGY

- **Acceptability judgment task**
- Thermometer scale (Experiment 1 also repeated with a 1–7 Likert scale)
- 4 "CP-phobic" English verbs: DEPEND, SPEAK, TALK, SUFFER
- 6 items per verb (based on Kim & Lu's 2024 materials)
- Fillers and standard items that cover the full spectrum of acceptability (roughly: A, B – grammatical, C – borderline, D, E – ungrammatical)
- Linear mixed-effects modeling

EXPERIMENTS

Experiment 1 ($N = 72$)

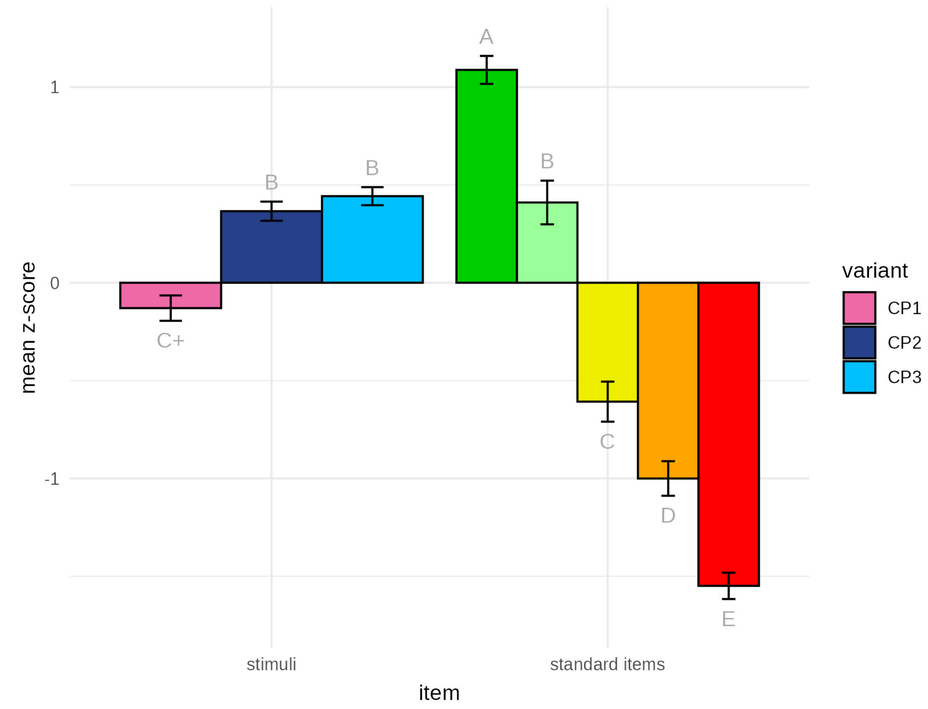
(1) The project **depends [** ...

(CP1) ... [$_{CP}$ **that** the new outsourced contractors will do their part],
[$_{PP}$ **on** an appropriate engineering design], **and**
[$_{PP}$ **on** the diligence of everybody involved]**]**.

(CP2) ... [$_{PP}$ **on** an appropriate engineering design],
[$_{CP}$ **that** the new outsourced contractors will do their part], **and**
[$_{PP}$ **on** the diligence of everybody involved]**]**.

(CP3) ... [$_{PP}$ **on** an appropriate engineering design],
[$_{PP}$ **on** the diligence of everybody involved], **and**
[$_{CP}$ **that** the new outsourced contractors will do their part]**]**.

- In all items, **each conjunct is exactly 12 syllables long**.



Results:

- Verb–CP **distance improves judgments**: CP1 vs. CP2 & CP3 ($p < .001$)
- The improvement **rapidly plateaus**: CP2 $\sim$ CP3 ($p = .12$)

Experiment 2 ($N = 77$)

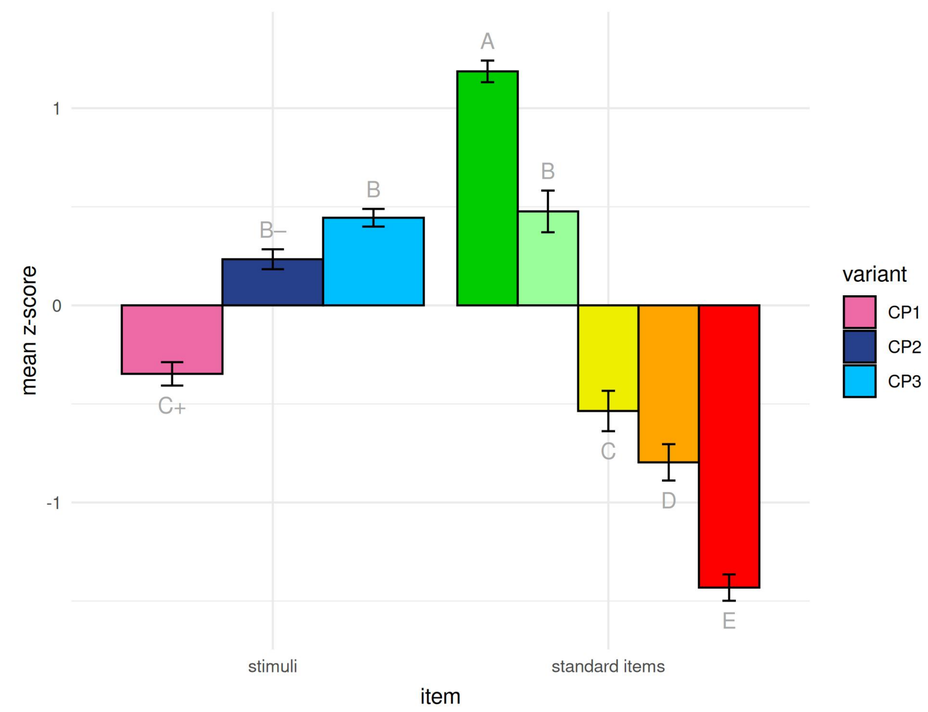
Length of conjuncts halved to 6 syllables each:

(2) The project **depends [** ...

(CP1) ... [$_{CP}$ **that** the morale stays high],
[$_{PP}$ **on** stakeholder input], **and**
[$_{PP}$ **on** timely decisions]**]**.

(CP2) ... [$_{PP}$ **on** stakeholder input],
[$_{CP}$ **that** the morale stays high], **and**
[$_{PP}$ **on** timely decisions]**]**.

(CP3) ... [$_{PP}$ **on** stakeholder input],
[$_{PP}$ **on** timely decisions], **and**
[$_{CP}$ **that** the morale stays high]**]**.



Results:

- **Distance still improves judgments**: CP1 vs. CP2 & CP3 ($p < .001$)
- The improvement **is detectable** at later positions: CP2 vs. CP3 ($p < .001$)

CONCLUSION

- **Linguistic representations decay**, as revealed in grammaticality illusions.
- Decay is **negatively accelerated: initially steep but levels off quickly.**

References

- Gibson, E. (1998). *Linguistic complexity: locality of syntactic dependencies.*
- Grodner, D. & Gibson, E. (2005). *Consequences of the Serial Nature of Linguistic Input for Sentential Complexity.*
- Kim, N. & Lu, J. (2024). *Coordination of Unlike Categories Creates Grammaticality Illusion.*
- Lewis, R. & Vasishth, S. (2005). *An Activation-Based Model of Sentence Processing as Skilled Memory Retrieval.*

Full paper is under review, but **available on demand**. **Talk/write to us:** b.sensekerci@uw.edu.pl adamp@ipipan.waw.pl kkus@uw.edu.pl aep@ipipan.waw.pl